

Question Number	Answer	Additional guidance	Mark
7(a)(i)	The correct answer is C two <i>A is incorrect because lignin gives waterproofing properties and can form spiral shapes</i> <i>B is incorrect because lignin gives waterproofing properties and can form spiral shapes</i> <i>D is incorrect because lignin does not add strength to the cell membranes</i>		(1)

Question Number	Answer	Additional guidance	Mark
7(a)(ii)	Similarity (1) • both contain {cellulose / pits / dead cells / secondary walls} • both {are hollow / do not contain cytoplasm} Difference (1) • sclerenchyma contain end walls whereas xylem do not	Ignore cell walls unqualified Ignore no nucleus Accept sclerenchyma are {pointed/closed} at each end whereas xylem are {not / open at each end} Accept xylem contain tracheids whereas sclerenchyma do not	(2)

Question Number	Answer	Additional guidance	Mark
7(b)	An explanation that makes reference to the following: • hemp is renewable / more can be grown / will not run out (1) • available to future generations (1)		(2)

Question Number	Answer	Additional guidance	Mark
7(c)(i)	• calculation of mean yearly increase (1) • calculation of total volume in 2020 (1) • correct answer for total volume in 2020 with units (1)	Example of calculation $(84,600,000,000 - 17,000,000,000) \div 11 = 6145454545.45$ $(6145454545.45 \times 20) = 1229090909 + 17,000,000,000 = 1.39 \times 10^{11} / 1.4 \times 10^{11} / 139909 \times 10^6$ $= 1.39 \times 10^{11} \text{ dm}^3 / 1.4 \times 10^{11} \text{ dm}^3 / 139909 \times 10^6 \text{ dm}^3$ Correct answer with no working gains full marks	(3)

Question Number	Answer
*7(c)(ii)	Answers will be credited according to candidate's deployment of knowledge and understanding of the material in relation to the qualities and skills outlined in the generic mark scheme. The indicative content below is not prescriptive, and candidates are not required to include all the material indicated as relevant. Additional content included in the response must be scientific and relevant. Advantages • indication of sustainability in the context of biofuel production e.g. more plants can be grown, renewable • available for future generations • consideration of carbon neutral or removal of carbon dioxide from atmosphere (by photosynthesis) as plant is growing • consideration of lower carbon emissions than fossil fuels • linkage to greenhouse effect / global warming • some biofuels require limited resources • hemp produces the least carbon dioxide emissions of all fuels • hemp and sugar beet require the least water • hemp and soybean require fewer fertilisers • hemp requires fewer fertilisers • hemp requires fewest resources • consideration of reduced water requirements for producing biofuels than food production • hemp requires fewest resources so lowest production costs • bioethanols require fewer resources so lower production costs Disadvantages • rapeseed and soybean production require high levels of water • rapeseed and sugar beet production require high levels of fertilisers • rapeseed, soybean and sugar beet production require high levels of pesticides • rapeseed requires highest level of resources • consideration of effect of fertilisers on ecosystem

		• consideration of effect of pesticides on ecosystem • disadvantage of reduced land available for food production • disadvantage of reduced water available for food production • disadvantage of reduced water available for population • cost of providing resources • biodiesels require more resources so higher production costs • supported conclusion of which bioethanol fuel would be most sustainable	(6)
			Additional guidance
Level 0	0	No awardable content	
Level 1	1-2	Limited scientific judgement made with a few strengths / weaknesses identified. A conclusion may be attempted, demonstrating isolated elements of biological knowledge and understanding but with limited evidence to support the judgment being made.	simple description of trends shown in the graph / table linked to basic advantages or disadvantages = lower mark simple description of trends shown in the graph / table linked to basic advantages AND disadvantages = higher mark
Level 2	3-4	A scientific judgement is made through the application of relevant evidence, with strengths and weaknesses identified. A conclusion is made, demonstrating linkages to elements of biological knowledge and understanding, with occasional evidence to support the judgement being made.	All level 1 plus sustainability should be considered to access level 2 conclusion of which biofuel would be best attempted linked to the information given for lower mark linked to the farmer OR population OR ecosystem for higher mark
Level 3	5-6	A scientific judgement is made which is supported throughout by sustained application of relevant evidence from the analysis and interpretation of the scientific	all level 2 plus consideration to effect on two from farmer OR population OR ecosystem lower mark

	information. A conclusion is made, demonstrating sustained linkages to biological knowledge and understanding with evidence to support the judgement being made.	consideration to effect on three from farmer OR population OR ecosystem higher mark
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